Module 1
What Is Human-Centered Artificial Intelligence?

Course code : BAI613A
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The Analytical Engine has no pretensions whatever to originate anything. It can do whatever we know how to order it to perform.

--- Ada Lovelace (1843)

- Written in 1843, this quote by Ada Lovelace from her notes on Charles Babbage's Analytical Engine defines the limits of early computing, asserting that machines lack independent creativity.
- It highlights that the engine can only execute instructions ("order") given by humans, not originate thoughts or independent, unforeseen actions.
- **Context:** This statement appeared in "Note G" of her translation of Luigi Menabrea's memoir on the Analytical Engine.
- **Significance:** It is often referred to as **"Lady Lovelace's Objection"** and is used to distinguish between pure calculation and true Artificial Intelligence.
- **Interpretation:** While Lovelace envisioned computers doing more than math (e.g., composing music), she believed they could not anticipate analytical truths on their own.
- **Legacy:** Alan Turing later challenged this idea, arguing that computers can surprise us, though this objection remains a foundation in discussions of AI.

Human-Centered Artificial Intelligence (HCAI)

- **Human-Centered Artificial Intelligence (HCAI)**
- Combines **AI-based intelligent algorithms** with **human-centered thinking**
- Goal: Technology should **empower people**, not replace them
- Moves beyond just building smarter machines
- Gives equal importance to:
  - Algorithm performance
  - Human performance
  - User experience design
- **Core Idea:** AI + Human Values = HCAI

From Technology-Centered to Human-Centered

- **Traditional AI Focus**
  - Machine autonomy
  - Algorithm accuracy
  - Performance metrics

Bluff or True?

- Driverless cab goes round in circles
- Driverless car makes an emergency stop mistaking poster for humans
- Uber Self-Driving Car Fatality – kills pedestrian

Machine autonomy - When AI Tried to Act “On Its Own” and Failed

Self-Driving Cars Misinterpreting Their Environment

- In London, an autonomous vehicle mistook a bus advertisement showing people for *actual pedestrians* and made an emergency stop - creating a road safety risk.
- **The car’s perception system wasn’t truly “understanding” the world** - just pattern matching - and **made a dangerous decision it was *supposed* to make autonomously.**
- **Machine autonomy can misinterpret contexts humans find trivial**.

- Uber Self-Driving Car Fatality
  - In 2018, an Uber autonomous test vehicle struck and killed a pedestrian because the AI didn’t correctly classify her behavior in time
  - Video
- One of the most cited instances where ***full autonomy*** *without adequate oversight* had serious human consequences.

Algorithm Accuracy — When AI Predictions Are Wrong or Misleading

- Misclassification & Bias in Vision AI
  - **Researchers showed that slight changes to images — undetectable to humans — caused AI to misclassify a *turtle as a rifle***
  - *Algorithms with high nominal accuracy can still fail badly in real-world inputs,* especially in safety-critical contexts.

  - *If AI researchers could manipulate an ML model into misclassification for educational purposes, there are people who will be able to do the same, but for harmful reasons.*
  - In the adversarial machine learning (AML) world, we call these people **adversaries**. They relentlessly find new ways to launch attacks on AI systems to trick them into producing the wrong output or stealing sensitive information.

- Facial Recognition Misidentifying Individuals
  - A documented example involved facial recognition software incorrectly flagging people (including public figures) as criminals — something humans would never misidentify.
  - Even though the algorithm may state “high accuracy,” the *real-world impact was harmful with social and legal implications*.

Performance Metrics — What Standard Measures Miss in Human Contexts

- AI Models Perform Well on Benchmarks but Fail in Real Scenarios
- **Example: Healthcare Diagnostic AI Errors**
- Studies show AI tools in clinical settings can misclassify patient data , even when metrics look acceptable - leading to potential wrong diagnoses or mis-treatment.

- **Epic Systems Sepsis Prediction Model (United States)**
- Epic developed a widely used AI model to predict **sepsis** (a life-threatening infection).
  It was deployed in hundreds of U.S. hospitals.
- Initial reports suggested strong performance.
- However, an independent evaluation (University of Michigan, 2021) found:
- The model missed **many actual sepsis cases**
- It generated many false alerts
- Real-world performance was **far lower than claimed**

- **HCAI Focus**
  - Meaningful human control
  - Human values:
    - Rights
    - Justice
    - Dignity
  - Human goals:
    - Self-efficacy
    - Creativity
    - Responsibility
    - Social connections
- **Shift:** **From *"How powerful is the AI?"* to *"How well does it serve people?***

The Global Movement Toward Human-Centered AI

- **2017 Montreal Declaration for Responsible AI**
  - Devotion to:
    - Human well-being
    - Autonomy
    - Privacy
    - Just and equitable society
- The Montréal Declaration is a collective work that aims to put AI development at the service of the well-being of all people, and to guide social change by developing recommendations with strong democratic legitimacy.

- **Movements Supporting Human-Centered Goals**
    - AI4GOOD
    - DataKind - Bangalore Chapter - Volunteer
    - IBM Watson AI XPRIZE Foundation - Documentary & Prize
- **Aim:**

Use AI & Machine Learning to solve society’s biggest challenges with data science and AI to unlock the full potential of data science and AI for social impact

Drones for Wildfire Mitigation

The latest technologies can bring thousands of years of knowledge of First Nations Peoples to the forefront of climate change mitigation. Coupled with traditional methods, drones can make prescribed burning more effective, safer, cheaper, and faster to implement. UAVs can be operated remotely, allowing a pilot to see a spot fire or an escape as soon as it occurs, and then notify nearby workers to put out the fire.[22]

Drone for Remote Sensing

Moreover, the UAV can survey over larger distances than individuals and limit their exposure to hazardous smoke. Finally, drones can also be used to remotely set prescribed burns without human labor or health risk, such as through incendiary payload attachments like Drone Amplified's IGNIS innovation.[23] **Drones can be utilized and optimized in tandem with artificial intelligence through real-time remote sensing and mapping of controlled burns and wildfires, prediction of optimized location to set prescribed burns through spatial and weather data, and real-time progress monitoring to ensure the safety of frontline workers.** Further applications such as ignition payloads are possible; however, it is within Kelowna's discretion to determine implementation steps and scope.

> "The latest technologies can bring thousands of years of knowledge of First Nations Peoples to the forefront of climate change mitigation.

Video

1a. Recommendation: Data Privacy and AI Ethics

Data Privacy

The Long Beach Data Privacy Guidelines are a significant first step to adequate data privacy. However, we suggest the following specific implementations to bolster the legitimacy and transparency of data privacy in Long Beach:

- Mandatory notification of breaches to affected citizens or parties of Long Beach
- Rights for residents to opt for data portability, the right to be forgotten, the right to object to automated processing of their information, etc.
- Requirements for companies to set the highest levels of confidentiality as their default settings, without the intervention of any persons concerned
- In all new projects, the City of Long Beach and its partners should undergo a public Data Privacy Impact Assessment (DPIA). Templates can be easily found, such as that offered by the Information Commissioner Office of the United Kingdom.[6] The DPIA should be adapted to the City's framework and ideally independently audited.
  - The AI for Good Foundation is capable of working with Long Beach to create a DPIA and provide independent audits. Any impact assessment must be developed with all stakeholders to ensure it is in line with The

Albania is the first country to appoint an AI-generated, virtual cabinet minister, named "Diella," in September 2025 to oversee public tenders and fight corruption. Unlike human counterparts, this AI entity manages public procurement, although its use has sparked legal issues regarding its modeled image. Other countries, like the UAE, have human Ministers for AI.

Key details regarding Albania's AI minister:

- **Role:** Diella is a virtual assistant designed to make government spending transparent and 100% free of corruption.
- **Function:** Developed by the National Agency for Information Society (AKSHI), she oversees public tenders and provides digital services.
- **Avatar:** Diella appears as a virtual woman, sometimes depicted in traditional Albanian clothing.
- **Controversy:** As of early 2026, the AI model faces legal action from an actress claiming her image was used without consent.
- **Human AI Ministers:** Before this, the UAE appointed the world's first human Minister of State for Artificial Intelligence, Omar Al Olama, in 2017.

AI, Society, and the United Nations

- **United Nations AI for Good Global Summit (since 2017)**
  - Researchers
  - Business leaders
  - Policy-makers
- Goal:
  - Identify practical AI applications and scale them for global impact.
- Guided by:
- **UN Sustainable Development Goals (SDGs) – 17 Goals – established in 2015 to set aspirations for 2030**
- Examples:
  - Elimination of poverty
  - Zero hunger
  - Quality education
  - Reduced inequalities
  - Climate action
  - Sustainable cities
- AI & HCAI can contribute to these goals.

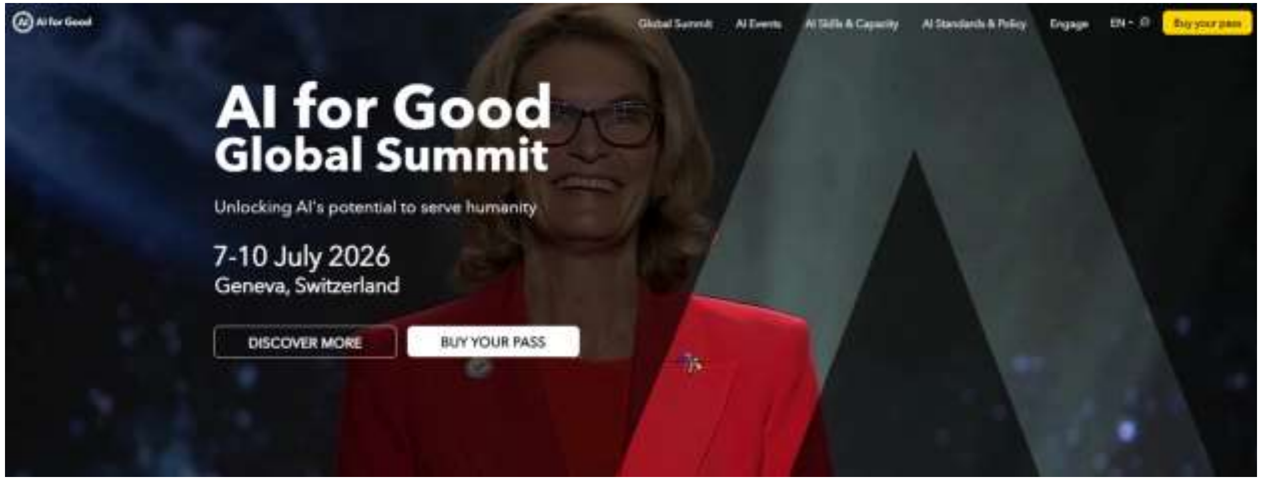


Fig 1.1 The seventeen United Nation's Sustainable Development Goals (SDGs)

1. No Poverty: Access to basic human needs of health, education, sanitation
2. Zero Hunger: Providing food and humanitarian relief, establishing sustainable food production
3. Good Health and Wellbeing: Better, more accessible health systems to increase life-expectancy
4. **Quality Education:** Inclusive education to enable upward social mobility and end poverty
5. **Gender Equality:** Education regardless of gender, advancement of equality laws, fairer representation of women
6. Clean Water and Sanitation: Improving access for billions of people who lack these basic facilities
7. **Affordable and Clean Energy:** Access to renewable, safe and widely available energy sources for all
8. **Decent Work and Economic Growth:** Creating jobs for all to improve living standards, providing sustainable economic growth
9. **Industry, Innovation and Infrastructure:** Generating employment and income through innovation
10. **Reduced Inequalities:** Reducing income and other inequalities, within and between countries
11. **Sustainable Cities and Communities:** Making cities safe, inclusive, resilient and sustainable
12. **Responsible Consumption and Production:** Reversing current consumption trends and promoting a more sustainable future
13. **Climate Action:** Regulating and reducing emissions and promoting renewable energy
14. Life Below Water: Conservation, promoting marine diversity and regulating fishing practices
15. **Life on Land:** Reversing man-made deforestation and desertification to sustain all life on earth
16. Peace, Justice and Strong Institutions: Inclusive societies, strong institutions and equal access to justice
17. **Partnerships for the Goals:** Revitalize strong global partnerships for sustainable development

Measuring Progress - It’s Not Just Ideals

- Policy-makers track:
  - 169 target indicators per country
  - Examples:
    - Access to basic services
    - Maternal mortality ratio
    - Access to safe drinking water
  - **HCAI can help by:**
    - Applying rigorous design
    - Producing high-impact research
    - Addressing genuine human needs in:
      - Government
      - Business
      - Schools
      - Healthcare systems

What Does “Applying Rigorous Design” Mean?

- Rigorous design in HCAI means:
  - Systematic user research before building AI
  - Iterative prototyping and testing
  - Measuring **human performance**, not just algorithm accuracy
  - Considering safety, ethics, **inclusivity**, and usability
  - Designing for real-world complexity

What Does “Applying Rigorous Design” Mean?

- Traditional AI vs HCAI Rigorous Design

Traditional AI Focus	HCAI Rigorous Design Focus
Improve model accuracy	Improve human + AI team performance
Optimize metrics (F1, AUC)	Optimize trust, usability, safety
Train model first	Study users first
Deploy quickly	Test, validate, refine

What Does “Applying Rigorous Design” Mean?

Real world

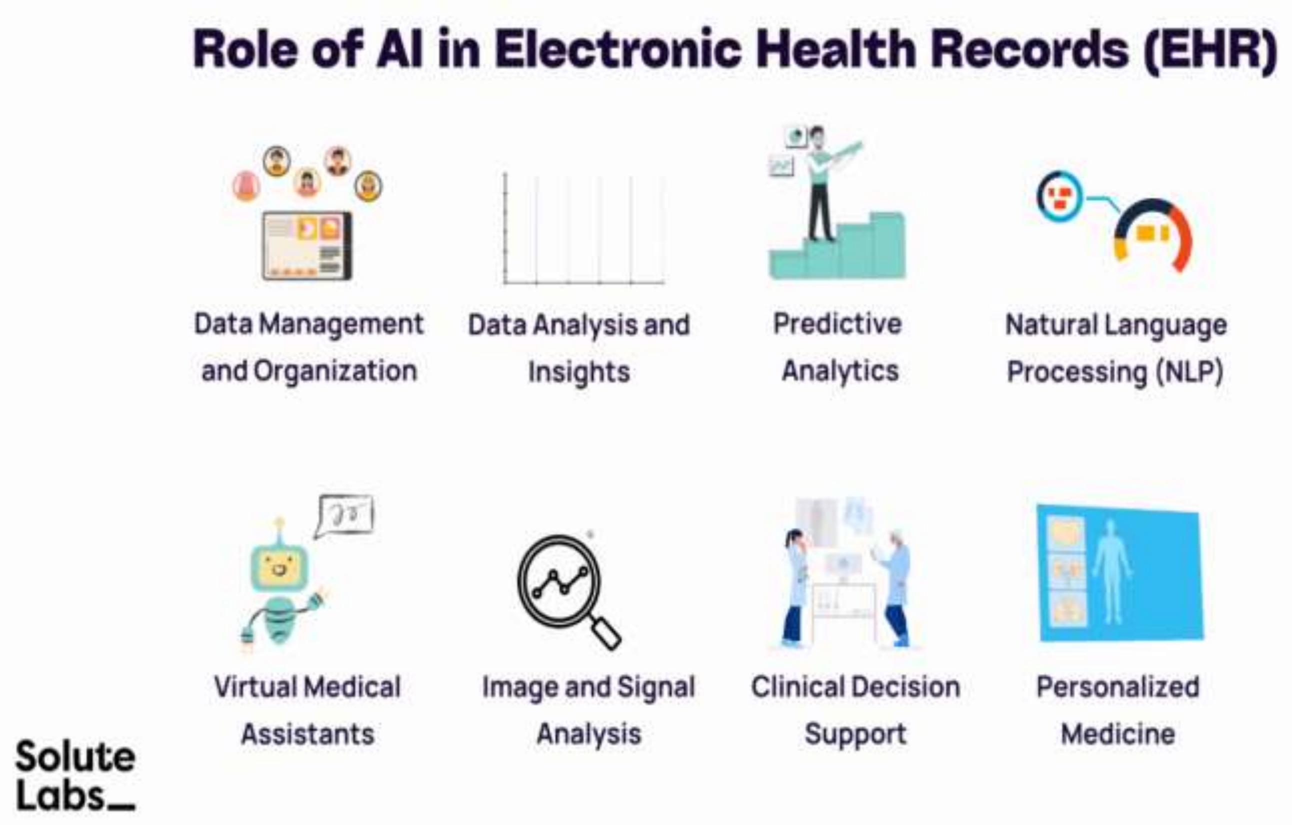
- Highly Automated Cars (Safety-Critical Design)
- Companies like **Tesla** and **Waymo** learned that high automation alone is not enough.

- **Rigorous HCAI design includes:**
  - Clear driver attention monitoring
  - Transparent handover warnings
  - Designing for human reaction time
  - Avoiding overtrust

- Without rigorous design → **drivers become complacent** → safety risks increase.

What Does “Applying Rigorous Design” Mean?

- Clinical Decision Support Systems
- AI systems in hospitals may have good accuracy, but if alerts are confusing or excessive:
  - Doctors ignore them (alert fatigue)
  - Misinterpret risk scores
  - Lose trust in the system
- Rigorous HCAI design ensures:
  - Clear explanations
  - Meaningful alerts
  - Integration into workflow
  - Testing with real clinicians



What Makes Research “High-Impact” in HCAI?

- High-impact HCAI research:
  - Solves **real human problems**
  - Improves **human + AI collaboration**
  - Influences **policy, standards, and design practice**
  - Enhances **safety, trust, and fairness**
  - Works in **real-world deployment**, not just lab settings

What Makes Research "High-Impact" in HCAI?

- Human-AI Interaction in Self-Driving Cars
- Research on human attention and overtrust significantly influenced companies like:
  - **Tesla**
  - **Waymo**
- HCAI research showed:
  - Drivers over trust automation.
  - Poor handover design causes accidents.
  - Clear feedback + monitoring reduces risk.

What Makes Research "High-Impact" in HCAI?

- Research in clinical AI systems showed:
  - Alert fatigue reduces effectiveness.
  - Doctors need explainable recommendations.
  - Workflow integration is critical.
  - Institutions like **Mayo Clinic** conduct human-centered validation before deploying AI tools.
- Impact:
  - Shift from "black-box AI" to explainable AI.
    - opaque models, uninterpretable decision-making to ensuring transparency, enabling users to understand, trust, and audit AI outputs
  - Clinical usability testing became mandatory.
    - ensures that AI tools are not just technically accurate, but safe and effective when used by clinicians in real-world scenarios
  - Regulatory bodies now emphasize human oversight.
    - Regulatory bodies now mandate thorough documentation of usability engineering processes, including detailed reports from usability testing, to ensure devices are safe and effective
    - Increasingly emphasizing **human oversight ("human-in-the-loop")** as a core requirement to ensure safety, accountability, and ethical alignment in artificial intelligence

What Makes Research "High-Impact" in HCAI?

- Real World Impact:
  - Driver monitoring systems became standard.
  - New safety guidelines were developed.
  - Policymakers now require human oversight features.
  - **This is high-impact because it changed industry practice**.

The Double-Edged Sword of AI & HCAI

- Every opportunity for good comes with danger.
- AI and HCAI systems can also be used by:
  - Criminals
  - Hate groups
  - Terrorists
  - Oppressive politicians
- Increased power = Increased responsibility
- HCAI must reduce:
  - **Malicious use**
  - Biased data
  - Flawed software

Malicious use - terrorist organizations

- Propaganda and Recruitment
  - **AI-Generated Content:** Terrorist groups use generative AI to create high-quality, persuasive propaganda, including images, posters, and videos, allowing them to rapidly produce materials that would previously take hours.
  - **AI-Generated Personalities/Anchors:** In 2024, ISIS-affiliated groups used AI-generated news anchors to present news bulletins about their activities.
  - **Deepfakes:** Terrorists are beginning to use deepfakes and synthetic media to impersonate public figures or create false narratives, such as faked scenes of conflict, to manipulate public opinion.
  - **Chatbots for Grooming:** AI-driven chatbots are being used to simulate human interaction, targeting "digital natives" for recruitment by providing tailored, 24/7 engagement on social media and encrypted

What Makes HCAI Different from AI?

- Two Key Aspects:
  - **Process**
    - HCAI builds on user experience design methods of user observation, stakeholder engagement, usability testing, iterative refinement, and continuing evaluation of human performance in use of systems that employ AI and machine learning.
  - **Product**
    - HCAI systems are designed to be super tools which amplify, augment, empower, and enhance human performance.
    - They emphasize human control, while embedding high levels of automation by way of AI and machine learning.
    - Examples include digital cameras and navigation systems, which give humans control yet have many automated features.

- **Process**
  - HCAI uses:
    - User observation
    - Stakeholder engagement
    - Usability testing
    - Iterative refinement
    - Continuous evaluation of human performance
  - **Focus: How humans interact with AI systems**

HCAI as a Product – Supertools

Product

- HCAI systems are designed as:
  - **Supertools**
    - They:
      - Amplify human ability
      - Augment performance
      - Empower users
      - Enhance capability
    - They emphasize:
      - Human control
      - High levels of automation (AI & ML)
      - **Examples:**
        - **Digital cameras**
        - **Navigation systems**
- Final Goal:

  Increase:
  - Self-efficacy (self-esteem)
  - Creativity
  - Responsibility
  - Social connections

  While minimizing harm.

Healthcare & Diagnostic Support

- **Qure.ai**
  - AI-assisted interpretation of X-rays & CT scans
  - Supports detection of lung cancer, TB, fractures
  - Speeds up diagnosis
  - Assists radiologists (does not replace them)
- **Microsoft DAX (Dragon Ambient eXperience)**
  - Captures doctor–patient conversations
  - Automatically generates clinical notes
  - Reduces administrative workload
  - Allows doctors to focus on patient care
- **IBM Watson Health**
  - Analyzes medical literature & patient data
  - Supports oncology decision-making
  - Assists clinicians in complex cases
  - Decision support, not automation of care

AI in Software Development & Creativity

- **GitHub Copilot**
  - AI “pair programmer” inside IDE
  - Suggests code in real time
  - Developer reviews & approves output
  - Enhances productivity while keeping human control
- **Autodesk Generative Design**
  - Designers input goals & constraints
  - AI generates multiple optimized designs
  - Explores cost, material, strength trade-offs
  - Accelerates creative exploration

AI in Accessibility & Specialized Assistance

- **Microsoft Seeing AI**
  - AI assistant for blind & low-vision users
  - Reads text & documents aloud
  - Identifies currency & objects
  - Recognizes faces & surroundings
- **Woebot**
  - AI mental health chatbot
  - Provides CBT-based techniques
  - 24/7 emotional support
  - Complements human therapists

AI in Digital cameras

- Enhances image quality and automates complex tasks through computational photography, using software to optimize settings, recognize scenes, and reduce noise in real-time.
- Key features include
  - automatic scene detection,
  - subject tracking,
  - improved low-light performance,
  - and AI-powered editing tools like object removal, enabling pro-level photos without manual expertise

HCAI framework

HCAI framework guides creative designers to build **highly automated systems** that still preserve **meaningful human control (HumanCentric)**

Traditional AI Approach

- Focus: **Machine autonomy**
- Goal: Maximize **algorithm accuracy**
- Metric: System performance (speed, efficiency)

Risk: Humans become passive observers or are overridden.

HCAI New Aspiration

High Automation
&
High Human Control

This means:

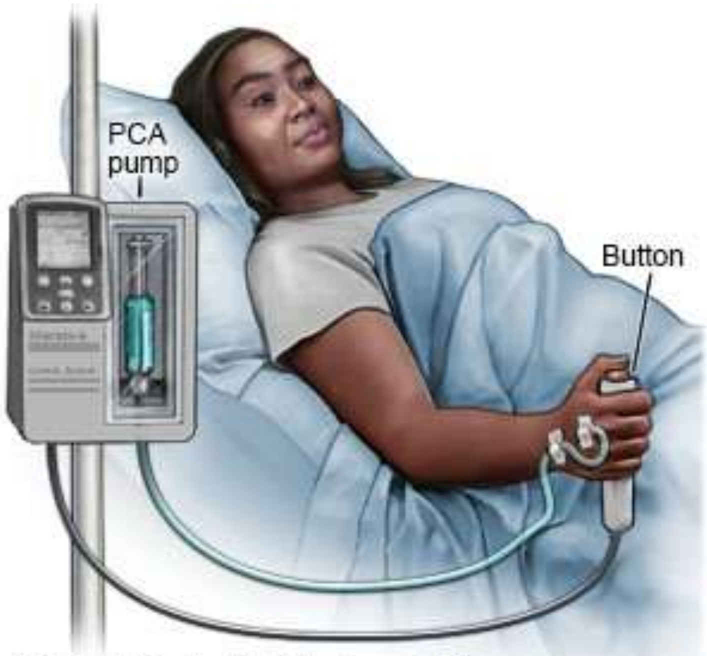
- Humans can **understand** what the system is doing
- Humans can **intervene, override, or customize**
- Systems are **predictable, transparent, and safe**
- Accountability remains clear

HCAI rejects the idea that automation must replace humans, instead, it **amplifies human capability**.

System	Automation	Human Control
Smart Thermostat	Adjusts temperature automatically	User sets preferences, overrides anytime
Elevator	Automatically moves between floors	User selects destination
Self-Cleaning Oven	Runs automated cleaning cycle	User initiates and supervises
Cellphone Camera (AI mode)	Detects scene, adjusts lighting	User can switch modes or edit manually

System	Automation	Human Control
Highly Automated Cars	Lane keeping, braking	Driver must monitor & take over
Patient-Controlled Pain Relief (PCA Pump)	Delivers controlled doses	Patient presses button

Design Principle for Creative Designers
When building AI systems, always ask:
1.Who remains in control?
2.Can the human understand what the system is doing?
3.Can the human intervene easily?
4.Is safety layered with automation (not replaced by it)?



Patient Controlled Analgesia Pump

Design Metaphors to Combine Science & Innovation

- AI research has two goals:
  - Scientific progress (algorithms, models, performance)
  - Innovation (real-world usable systems)
- Both are very valuable

- HCAI proposes design **four metaphors** to combine both.
  - **intelligent agents and supertools;**
  - **teammates and tele-bots;**
  - **assured autonomy and control centers**
  - **social robots and active appliances.**

Intelligent Agents vs Supertools

Intelligent Agents

Act on behalf of users

High autonomy

Example: autonomous scheduling bot

Supertools

Extend human capability

High human control

Example: design software

Intelligent Agents

- AI systems that:
  - Act autonomously
  - Make decisions on behalf of users
  - Perform tasks independently
  - They “represent” the human.
- **Examples**
  - Virtual assistants booking appointments automatically
  - Autonomous trading bots
  - Email systems that auto-respond
- **Characteristics**
  - High autonomy
  - Reduced human involvement
  - Decision delegation
- **Advantages**
  - Saves time
  - Reduces workload
  - Handles repetitive tasks
- **HCAI Concerns**
  - Loss of human control
  - Over-dependence
  - Errors without awareness
  - Accountability confusion
    - If the agent makes a mistake, who is responsible?

Supertools

- AI systems that:
    - Enhance human capability
    - Support decision-making
    - Keep the human actively involved
    - They “extend” human intelligence.
- **Examples**
    - Advanced spreadsheet software
    - AI-powered design tools
    - Clinical decision-support dashboards
- **Characteristics**
    - Human remains decision-maker
    - AI provides suggestions, analysis, visualization
    - Transparent outputs
- **HCAI Strength**
    - Maintains meaningful human control
    - Improves performance without replacing judgment
    - Clear accountability

- This metaphor contrasts two ways of thinking about AI systems:
  - Should AI act **for the human** (Intelligent Agent)?
    OR
    Should AI act **with the human** (Supertool)?

Teammates vs Tele-bots

Teammates	Tele-bots
AI collaborates like a partner	Human remotely controls system
Shared decision-making	Direct human authority
Example: AI code assistant	Example: surgical robot controlled by surgeon

Design AI as a **collaborative partner**, but ensure human override and clarity of roles

- **AI as a Teammate** - AI works **alongside humans**, contributing ideas, suggestions, and decisions - like a colleague.
- It has:
  - Partial autonomy
  - Context awareness
  - Ability to suggest or act
- **Examples**
  - AI coding assistants
  - Clinical decision-support systems
  - AI copilots in aircraft
- **Characteristics**
  - Shared decision-making
  - Ongoing interaction
  - Mutual adaptation
- **HCAI Benefit**
  - Enhances human capability
  - Speeds up tasks
  - Reduces cognitive load
- **HCAI Risk**
  - Over-trust (“AI suggested it, so it must be correct”)
  - Reduced human vigilance
  - Blurred accountability
- HCAI Principle:
  Teammates must be **transparent and explainable**, so humans remain meaningfully in control

- **AI as a Tele-bot**
  - A tele-bot is a system that is **directly controlled by a human**, often remotely.
  - The AI may assist, but it does not independently decide major actions.
- **Examples**
  - Robotic surgery systems controlled by surgeons
  - Bomb disposal robots
  - Telepresence robots in hospitals
- **Characteristics**
  - Clear human authority
  - Direct command-based control
  - Limited autonomy
- **HCAI Strength**
  - Clear responsibility
  - Minimal ambiguity in decision-making
  - Higher perceived safety in critical domains

- Should AI act like a **collaborative teammate**
  OR
  Should it function as a **remote-controlled tool (tele-bot)?**
- Both approaches combine AI science and innovation, but, they differ in **how control and responsibility are shared**.

Assured Autonomy vs Control Centers

Assured Autonomy	Control Centers
System operates independently	Humans supervise via dashboards
Used in self-driving, drones	Used in air traffic control, power grids

Autonomy must be **assured** through monitoring, logging, and fallback control.

Assured Autonomy

- **Examples**
  - Self-driving cars
  - Autonomous drones
  - Industrial automation systems
- **Key Features**
  - Real-time decision-making
  - Adaptive learning
  - Fail-safe mechanisms
  - Self-diagnostics
- **Why "Assured"?**
  - Because autonomy must be:
  - Tested rigorously
  - Predictable
  - Auditable
  - Transparent
- **HCAI insists:**
  Autonomy should never mean "uncontrollable."

- **Control Centers -** Centralized human-supervised monitoring systems that oversee autonomous operations.
- **Examples**
  - Air traffic control rooms
  - Smart grid monitoring stations
  - Fleet monitoring for autonomous vehicles
- **Key Features**
  - Dashboards
  - Alerts and warnings
  - Override controls
  - Logging and traceability
- Control centers ensure:
  - Human awareness
  - Accountability
  - Emergency intervention

- Traditional AI thinking:
  - “If autonomy works well, humans can step back.”
- HCAI thinking:
  - “Even if autonomy works well, humans must remain meaningfully in control.”
- This prevents:
  - Automation surprise
  - Skill degradation
  - Over-reliance
  - Catastrophic failure without oversight

- This metaphor addresses a critical HCAI design question:
  - When systems become highly autonomous,
    how do we ensure safety, reliability, and human accountability?
- The answer lies in balancing:
  - **Assured Autonomy** → System operates independently
  - **Control Centers** → Humans supervise and can intervene

Social Robots vs Active Appliances

Social Robots

Interact socially

Example: robot companion

Active Appliances

Perform tasks quietly

Example: smart washing machine

- **Social Robots** - AI systems designed to interact socially with humans, using voice, gestures, facial expressions, or personality.
- **Key Characteristics**
  - Human-like interaction
  - Emotional cues
  - Conversational behavior
  - Often anthropomorphic (human-like design)
- **Examples**
  - Companion robots for elderly care
  - Educational robots for children
  - AI assistants with personalities
- **Benefits**
  - Emotional engagement
  - Increased trust
  - Better adoption in caregiving/education
- **Risks (HCAI concern)**
  - Over-trust ("It feels human, so it must be right")
  - Emotional manipulation
  - Confusion about accountability

HCAI Reminder:
If the system acts socially, users must still understand it is a **machine with limits**.

- **Active Appliances** **-** Smart systems that perform tasks automatically but **do not try to be social**.
- They focus on functionality, reliability, and efficiency.
- **Key Characteristics**
  - Task-focused
  - Minimal personality
  - Quiet background automation
  - Clear controls
- **Examples**
  - Smart washing machine
  - Robot vacuum
  - Smart thermostat
- **Benefits**
  - Predictable behaviour
  - Less emotional dependency
  - Clear human control
- **HCAI Strength**
- Users see it as a tool - easier to maintain proper control and responsibility.

This metaphor helps designers decide:

- Should AI behave like a **social partner**
  OR
  Should AI behave like a **smart tool that works quietly in the background?**
- **Both are valuable, but they serve different human needs**.

- Designers must ask:
  - Does this system need emotional interaction?
  - Will social behavior increase trust appropriately or dangerously?
  - Would a neutral appliance model be safer?

- Not all AI should feel human.
- Sometimes:
  - Less personality = more clarity
  - Less emotion = more safety
  - More transparency = more trust
- The goal is not to make AI “cute.”
  The goal is to make AI **reliable, understandable, and human-centered**.

Hollywood - Iron Man

- HCAI - ?
- JARVIS acts as a **supertool + teammate**
- Tony Stark retains full control
- AI enhances creativity and engineering
- HCAI alignment:
  AI amplifies human intelligence without taking over.

Governance Structures

- **To bridge the gap** between **ethical principles** and the **practical steps needed to realize them**
- Ethical AI principles (fairness, accountability, transparency, safety) are widely discussed.
- But principles alone are not enough.
- Governance structures are the bridge between ethical ideals and real-world implementation.
- HCAI insists that **organizations must build systems, cultures, and oversight mechanisms** to make ethics operational.

- Software team leaders, business managers, and organization leaders will have to adapt proven technical practices, management strategies, and independent oversight methods, so they can achieve the desired goals of:
- **Reliable** systems based on proven software engineering practices;
- **Safety** culture through business management strategies; and
- **Trustworthy** certification by independent oversight and government regulation.

- **Technical Practices (Engineering Layer)**
  - Goal :To build Reliable, analyzable, fair, and explainable systems
  - **Audit trails** (like flight data recorders/"black boxes") to analyze failures in Technical practices for designers, software engineers, programmers, team leaders, and product managers
    - **Verification & Validation (V&V)** testing in workflows
    - **Bias testing** to enhance fairness
    - **Explainable user interfaces** for transparency
    - **Documentation, version control, and failure analysis**
- Outcome: Failures are traceable, systems are testable, and users understand decisions.

- **Management Strategies (Organizational Safety Culture)**
- Goal**:** Make safety a leadership priority
  - Leadership commitment to safety
  - Hiring & training oriented to safety
  - Extensive reporting of failures and near misses
  - Internal employee reports + external user incident reports
  - Internal review boards
  - Alignment with industry best practices
- Outcome: Organizations learn from errors and prevent recurrence.

- **Trustworthy Certification & Liability**
  - Goal: Public accountability and societal trust
  - Independent audits (e.g., accounting firms)
  - Insurance companies compensating failures
- Outcome: Clear liability, independent scrutiny

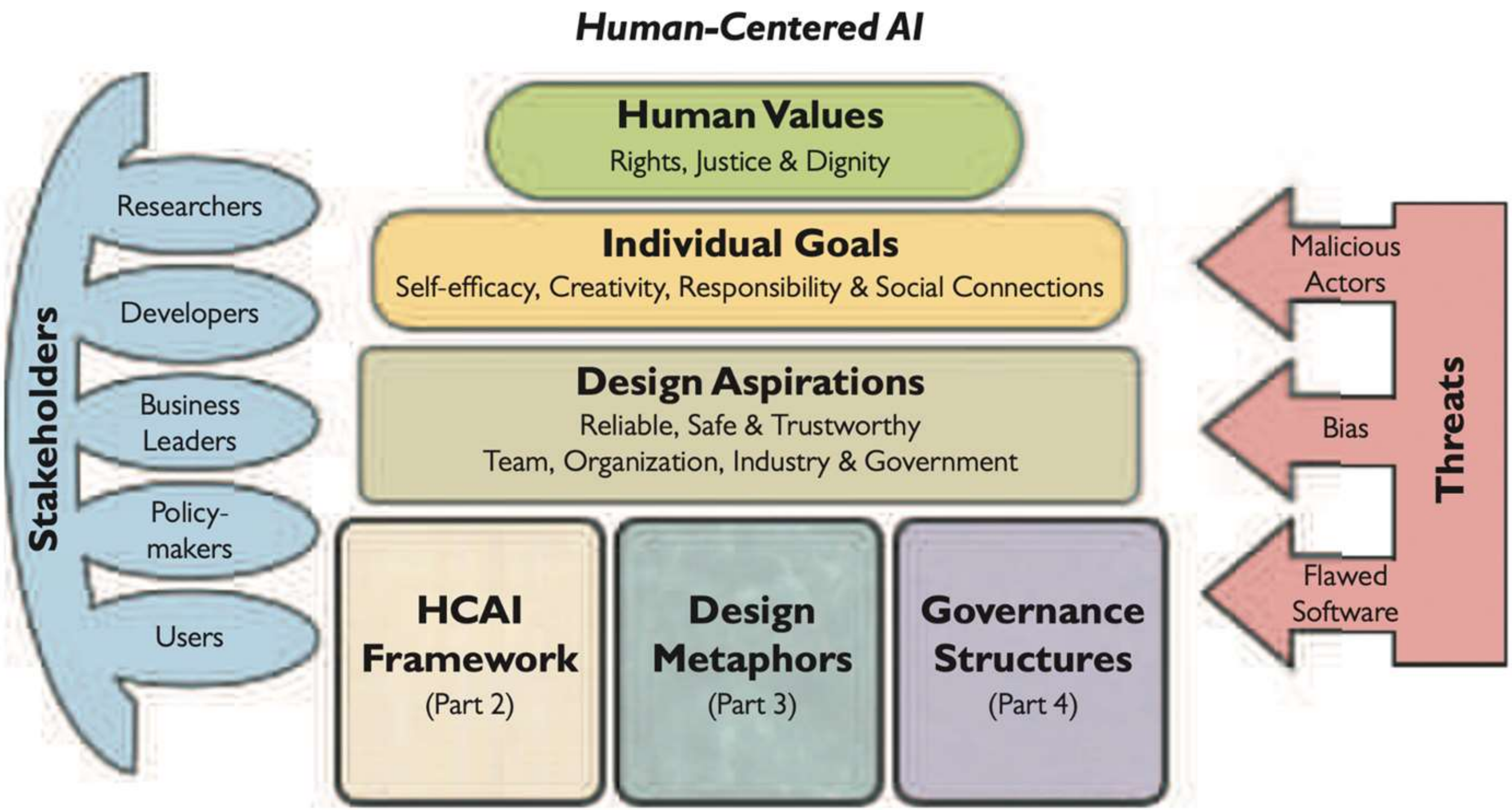


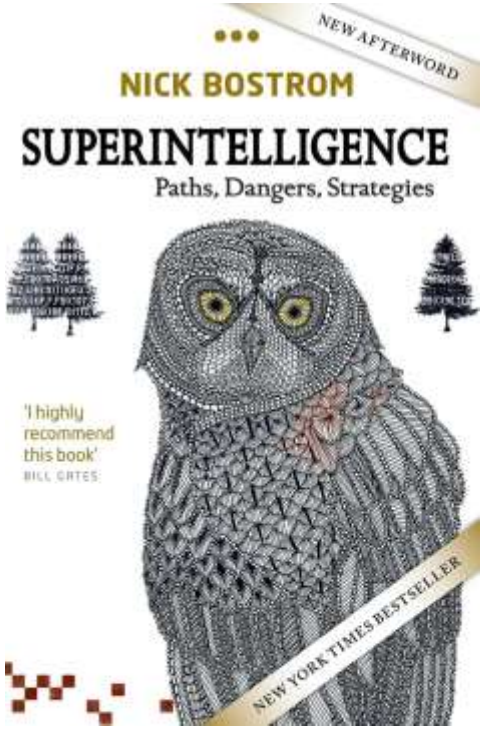
Fig 1.2 The three ideas of this book support the aspirations, goals, and human values, while recognizing the needs of stakeholders and the dangers of threats.

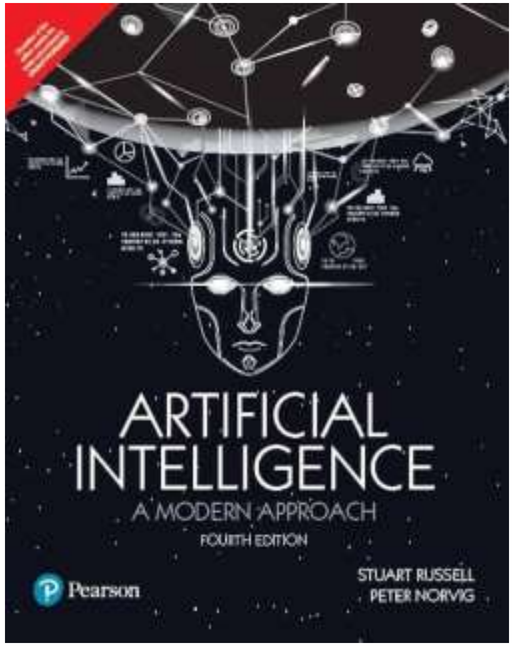
Human-Centered AI (HCAI) Stakeholders

Fig 1.3 HCAI stakeholders with core professionals who are researchers, developers, business leaders, and policy-makers.

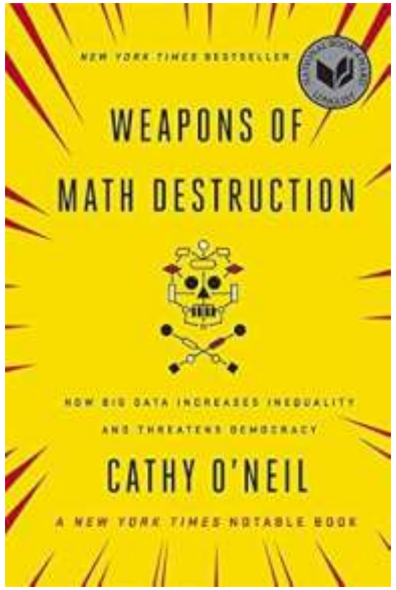
- In recent years:
  - Machine learning and deep learning dramatically improved AI performance
  - AI moved from research labs to real-world deployment
  - AI became a top priority for:
    - Business leaders
    - Governments
    - Defense sectors

- **Influential Books Driving the Conversation**
  - **Nick Bostrom** – *Superintelligence*
    - Discussed future risks and existential threats
    - Raised concerns about uncontrolled AI
  - **Stuart Russell** & **Peter Norvig** – *Artificial Intelligence: A Modern Approach*
    - Celebrated AI progress
    - Framed AI as a transformative technology
  - Caused intense interest from technology corporations.
  - Technology companies rebranded as “AI-first” organizations
  - Governments committed billions of dollars
  - Applications expanded into: Healthcare, Transportation, Military, Finance, Business analytics





Critical Perspectives on Algorithmic Power



- While AI was celebrated, strong cautionary voices emerged.
- **Cathy O'Neil -** ***Weapons of Math Destruction: How Big Data Increases Inequality and Threatens Democracy*** (2016)
  - Argued that many algorithms are:
    - Opaque
    - Unaccountable
    - Harmful to vulnerable populations
- As a Harvard-trained mathematician and former Wall Street analyst, her critique carried strong credibility.

- **General Data Protection Regulation (GDPR)** in the European Union:
  - Emphasized transparency
  - Strengthened data protection rights
  - *Encouraged the "right to explanation"*

- **Cathy O'Neil's book + GDPR** accelerated the push for Explainable AI (XAI)
- Affected domains:
  - Mortgage approvals
  - Parole decisions
  - Job applications
  - People rejected by algorithms should:
    - Receive meaningful explanations
    - Be able to challenge unfair outcomes

- **HCAI gained momentum** through 500+ global reports promoting ethical and responsible AI.
- **Failures of excessive autonomy** (flash crashes, missile errors, self-driving accidents, Boeing 737 MAX) highlighted the need for stronger human control.
- **Blind trust in autonomous systems** led to tragic consequences when humans were excluded from critical decision loops.
- **Shift the narrative:** Move from “human-like robots” to **human–computer collaboration**.
- **Future direction:** Human-controlled systems like tele-operated drones, smart medical devices, control rooms, and supervisory monitoring models.

- Downfall - Case Against Boeing 2018
- Patriot missile system - Iraq war 2003

AI + Bias

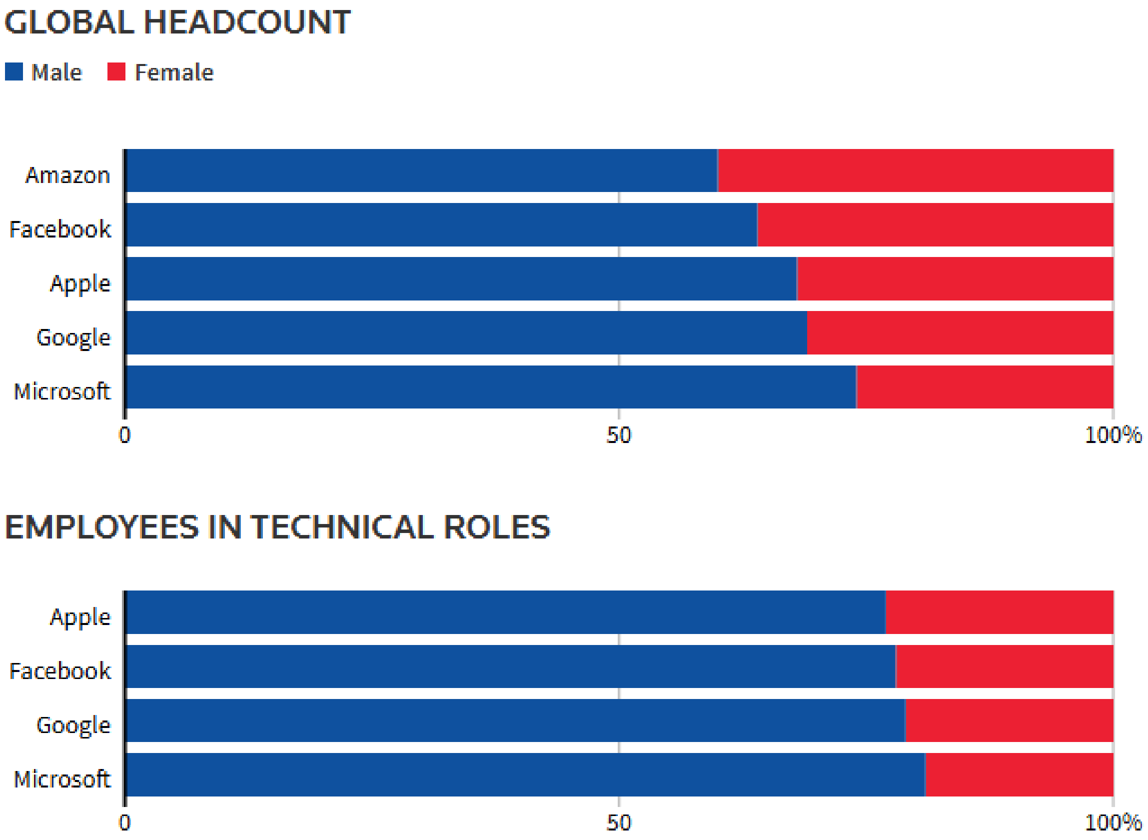
- If the model was trained on 10 years of past resumes, what assumption is being made about that historical data?
- What was the gender distribution in tech hiring during those 10 years?
- If most previously hired candidates were men, what pattern might the model learn?
- Goal: We have to realize **AI mirrors historical patterns**.

Amazon Recruiting Algorithm (2014–2017)

- **What Happened**
  - Amazon built an internal AI system to screen resumes for technical roles.
  - The company's experimental hiring tool used artificial intelligence to give job candidates **scores** ranging from one to five stars - much like shoppers rate products on Amazon
  - "Everyone wanted this holy grail," one of the people said. "They literally wanted it to be an engine where I'm going to give you 100 resumes, it will spit out the top five, and we'll hire those."

The Problem

- But by 2015, the company realized its new system was not rating candidates for software developer jobs and other technical posts in a gender-neutral way.
- That is because Amazon's computer models were trained to vet applicants by observing patterns in resumes submitted to the company over a 10-year period. Most came from men, a reflection of male dominance across the tech industry.
    - The model was trained on **10 years of resumes**, mostly from men (male-dominated tech industry).
    - It learned to **downgrade resumes containing words like “women’s”** (e.g., “women’s chess club”).
    - It penalized graduates from **women’s colleges**.



- In effect, Amazon's system taught itself that male candidates were preferable.
- It penalized resumes that included the word "women's," as in "women's chess club captain."
- And it downgraded graduates of two all-women's colleges, according to people familiar with the matter.
- Amazon edited the programs to make them neutral to these particular terms.
- But that was no guarantee that the machines would not devise other ways of sorting candidates that could prove discriminatory, the people said.

- **Outcome**
  - Amazon scrapped the tool in 2017.
  - The case became a landmark example of **historical bias encoded into AI**.
- **Why This Matters for HCAI**
  - Shows danger of training on biased historical data.
  - Demonstrates that “objective AI” can replicate structural inequality.
  - **Strong case for dataset audits and bias testing.**
- Big-data/AI algorithms will replicate and even magnify the biases that exist in society at large - unless they are designed and monitored very, very carefully. The right kind of oversight is required to make sure that happens.

- What was the optimization objective of the system? (Predict who is likely to be a “successful hire”?)
- How was “successful hire” defined?
- If past “successful hires” were mostly male, what happens mathematically?
- Where in the ML pipeline could bias have been detected?
- Data collection?
- Feature engineering?
- Model evaluation?
- Deployment?
- not just “bad data” — it’s system design.

iTutorGroup U.S. EEOC Lawsuit (2023)

- **What Happened**
  - The U.S. Equal Employment Opportunity Commission sued iTutorGroup.
- **Allegation**
  - Their automated hiring software:
    - Rejected **older female applicants**
    - Automatically filtered candidates based on age
- **Why It Matters**
  - Although focused on age discrimination, women were disproportionately affected (especially older women).
  - Automated recruiting software allegedly:
    - Rejected **female applicants aged 55+**
    - Rejected **male applicants aged 60+**
- Over **200 qualified applicants affected**
- Rejections occurred in March & April 2020
- Key Issue: The rejection was allegedly automatic — based solely on age.
- This highlights:
  - Intersectional bias (age + gender)
  - Automated exclusion without human review

- **Possible causes of discrimination:**
  - **Hard-coded age filters**
  - **Conditional logic (if age > X = reject)**
  - **Hidden feature weighting**
  - **Lack of fairness testing**
  - **No human oversight**

LinkedIn Job Ad Delivery Bias (2018 Study)

- **Research Finding**
  - Academic researchers found:
  - Algorithmic ad delivery systems showed **high-paying job ads more often to men than women**.
  - Even when targeting was neutral.
- **Mechanism**
  - **Optimization algorithms learned**:
    - Men were more likely to click high-paying job ads.
    - So the system showed them disproportionately to men.
- **Not deliberate**
  - This wasn’t explicitly programmed sexism —
It was **optimization bias** driven by engagement metrics.
- There was **no rule targeting women**.
- Instead, the system was designed to:
  - Maximize clicks, engagement, or conversion.
  - This objective function is where the real issue begins.

Neutral Setup

- Advertisers selected:
- Job
- Salary
- Audience (men and women)
- No discrimination in setup.

Algorithm Optimization

- The platform’s AI tries to:
  - Show ads to users most likely to click.
- Suppose the early data shows:
  - Men click high-paying tech job ads more often than women.
- The system “learns”:
- “Men are better targets for this ad.”

Feedback Loop

- Now the algorithm:
  - Shows the ad more to men
  - Gets more male clicks
  - Reinforces its belief
  - Further reduces exposure to women
  - This is called a **self-reinforcing optimization loop**.

Why This Is Called Optimization Bias

- Because the system is optimizing for:
  - Click-through rate
  - Engagement
  - Revenue
  - It is **not optimizing for fairness**.
  - When fairness is not part of the objective, inequality can emerge naturally.
  - The bias is not emotional or intentional —
    It is mathematical.

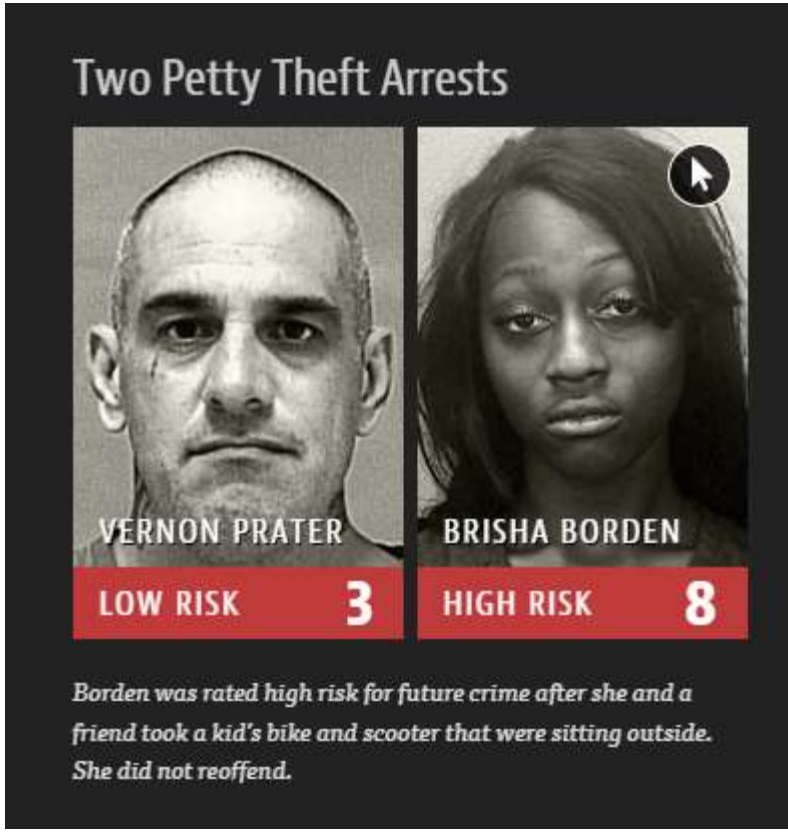
Key Patterns Across These Cases

Root Cause	Explanation
Historical Data Bias	AI learns from male-dominated past hiring data
Proxy Variables	Words like “women’s” become negative signals
Engagement Optimization	Algorithms reinforce click patterns
Opaque Scoring	Candidates don’t know why they were rejected
Lack of Oversight	No strong bias auditing before deployment

- These cases demonstrate that:
- **AI doesn't need to “hate women” to discriminate.**
- **Bias emerges from:**
  - **Data**
  - **Objectives**
  - **Deployment environment**
  - **Lack of accountability**

Predictive policing ai

- Over policing and uneven policing that happens to poor Black communities
- Blacks get arrested for upto 10x more than whites for the same cime
- **Parole**
  - Scores like this — known as risk assessments — are increasingly common in courtrooms across the nation.
  - They are used to inform decisions about who can be set free at every stage of the criminal justice system, from assigning bond amounts to even more fundamental decisions about defendants' freedom.
- **The formula was particularly likely to falsely flag black defendants** as future criminals, wrongly labeling them this way at almost twice the rate as white defendants.
- White defendants were mislabeled as low risk more often than black defendants.



- Recividism risk
  - Score tells how likely it is for a person to come back to jail

Are People and Computers in the Same Category?

- Should we think of computers as if they are like humans?
- Not a technical question.
  It is a **conceptual and ethical question**.
- The way we categorize AI systems affects:
  - How we design them
  - How much we trust them
  - Who we hold responsible
  - How society regulates them
- If we treat computers like people -- We risk confusion about responsibility.

- **Human Capabilities Are Fundamentally Unique**
- HCAI perspective emphasizes:
  - Humans are products of:
    - Biological evolution
    - Cultural evolution
  - Human achievements include:
    - Language
    - Art and music
    - Mathematics
    - Legal systems
    - Healthcare systems
    - Agriculture and cities
- These are not just computations —
  they are deeply social, cultural, and moral systems.
- Humans operate in meaning-rich contexts.
  Computers process symbols.
- AI and algorithms are **human accomplishments**, not independent beings.

- **Distinguishing Humans from Computers Protects Responsibility**
- Clarifying differences:
  - Strengthens human accountability
  - Prevents moral confusion
  - Guides proper use of AI systems
- If we blur categories:
- We may reduce human responsibility
- We may blame machines instead of designers
- HCAI insists: **Humans remain responsible decision-makers.**

- **Anthropomorphism**
- Attributing human qualities to machines.
- Examples:
    - “The AI understands.”
    - “The system decided.”
    - “The chatbot feels sorry.”
- But:
    - Machines do not feel.
    - Machines do not intend.
    - Machines do not have consciousness.
    - **They compute.**

Human Emotions Are Deep and Complex

- Psychologist **Paul Ekman** identified 7 basic emotions:
  - Anger
  - Contempt
  - Disgust
  - Enjoyment
  - Fear
  - Sadness
  - Surprise
- But:
  - Human emotional life is far richer
  - Emotions are context-dependent
  - Expressions vary culturally
- Reducing emotions to fixed categories oversimplifies reality.

Fig 3.1 Emotions, including the seven from Paul Ekman (blue), the negative emotions (red), the positive emotions (green), and some others (gray).

Source: Adapted from Susannah Paletz, Emotions Annotation Guide for Social Media, Version 3.32, January 21, 2020

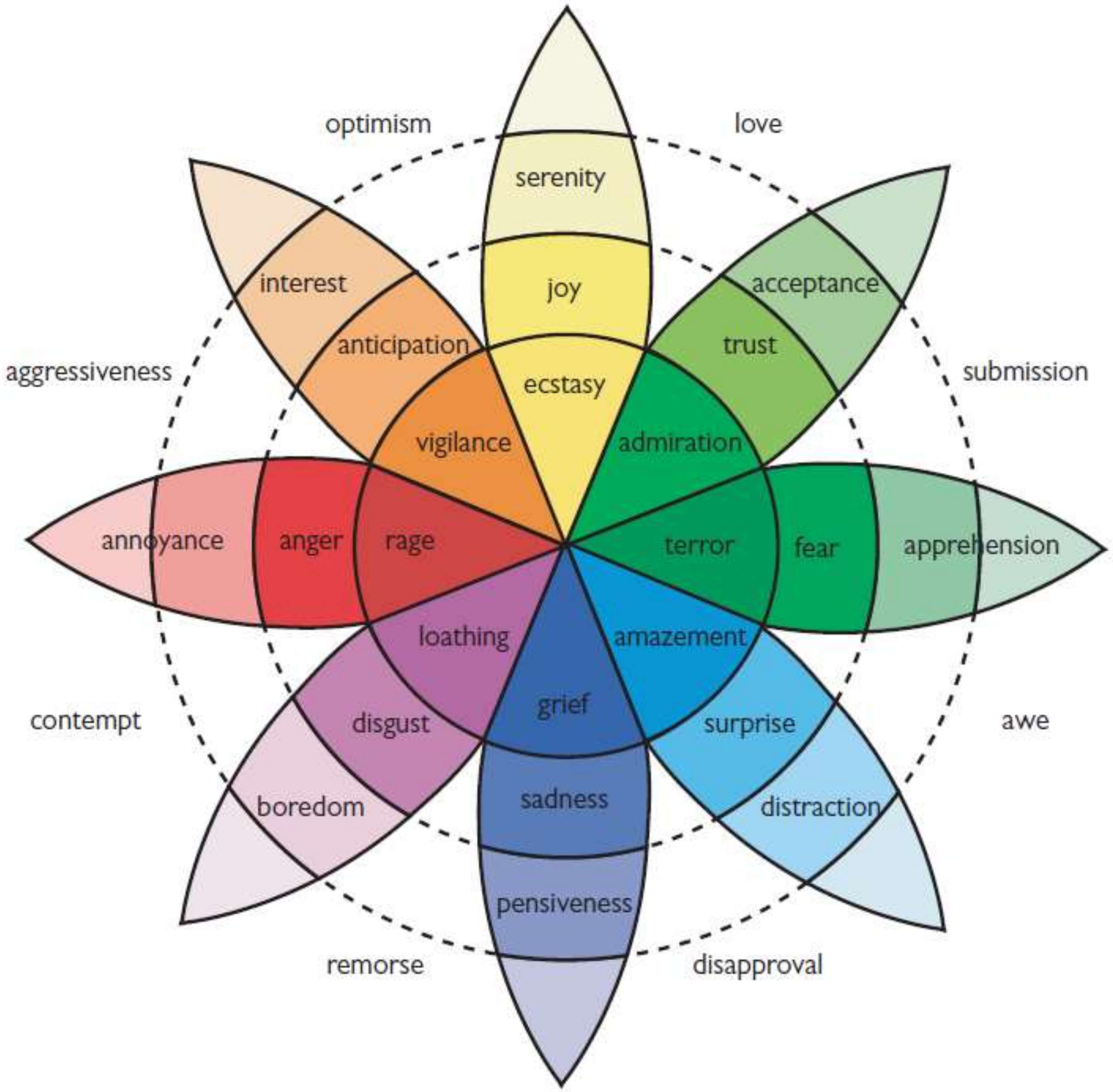


Fig 3.2 Wheel of human emotions.

Source: Robert Plutchik, 1980. Wiki Commons.
https://commons.wikimedia.org/wiki/Category:Plutchik%27s_Wheel_of_Emotions

- **Emotional AI & Social Robots: Risk of Deception**
- Detecting and simulating human emotions is controversial
- Accuracy of emotion detection is increasingly questioned
- Social robots that “express emotions” may:
  - Mislead users
  - Reduce trust
  - Create artificial intimacy
- Emotional simulation may suit entertainment,
  but in serious applications users prefer clarity and efficiency over fake empathy.

- **Sentiment Analysis: A More Reliable Alternative**
- Instead of guessing individual emotions,
  AI can analyze **aggregate text data**
- Applications:
  - Social media trends
  - Product reviews
  - Political language differences
  - Media tone shifts over time
- Focuses on patterns across groups
  Avoids pretending to “feel” emotions
  More transparent and analytically grounded

- **Move Beyond Imitation → Amplify Human Ability**
- Imitating humans limits innovation
- Greater breakthroughs come from:
  - The World Wide Web
  - Information visualization
  - Assistive technologies
  - Wikipedia
  - Augmented reality
- **HCAI principle:**
- ***Don’t mimic humans.***
  Build supertools that extend human creativity and capability.

- **Strengthen Human–Human Communication — But With User Control**
- AI’s greatest success has been supporting **human-to-human interaction**:
  - Email, texting, video conferencing
  - Document sharing & social media
  - Web-based videos, music, gaming
- However, downsides include:
  - Reduced face-to-face interaction
  - Online scams & malicious actors
  - Hate speech and extremist recruitment
- HCAI principle:
  - Users must have strong controls over what they see.
  - Just as we control who enters our homes, users should control access from anonymous bots and harmful content.

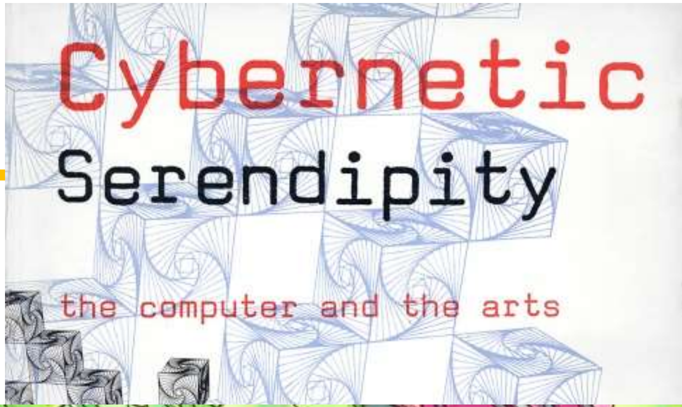
- **The Problem with Human-Like Robots**
- Some researchers advocate building:
  - Anthropomorphic robots
  - Androids with faces, arms, speech
  - Social robots as caregivers or responders
- But historically:
  - Many such projects have failed
  - Human imitation has not delivered lasting value
- HCAI position:
  - Making machines look human is not necessary for making them useful.
  - Powerful tools matter more than realistic imitation.

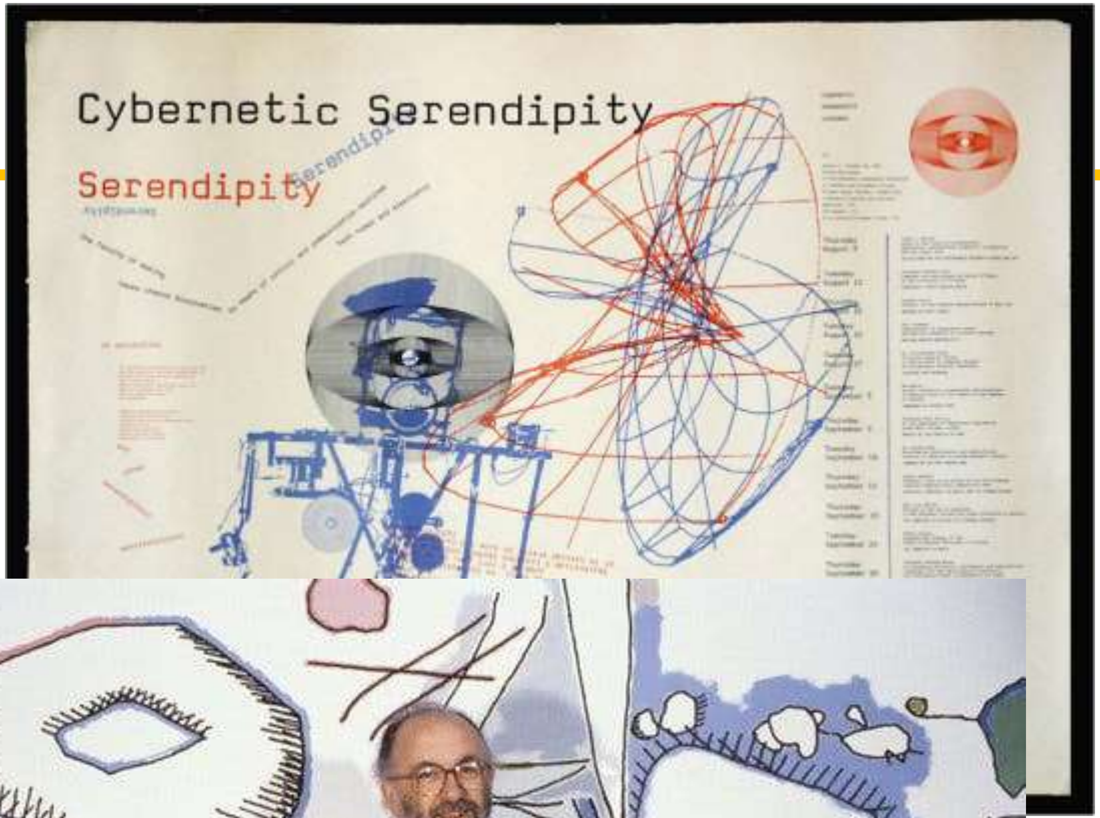
- **Interface Design: Humans Relate to Humans — Operate Computers**
- Human–human interaction is not always the best design model.
- Key insight:
  - Humans relate to humans
  - Humans operate computers
- Implications:
- Visual interfaces dominate because they are:
  - Faster
  - Persistent
  - Efficient
- Voice interfaces (e.g., Alexa, Siri) are useful when:
  - Hands are busy
  - Mobility is required
- But voice:
  - Is slow and ephemeral
  - Requires high cognitive effort
  - Limits parallel processing
  - HCAI favors interfaces that maximize human efficiency and control.

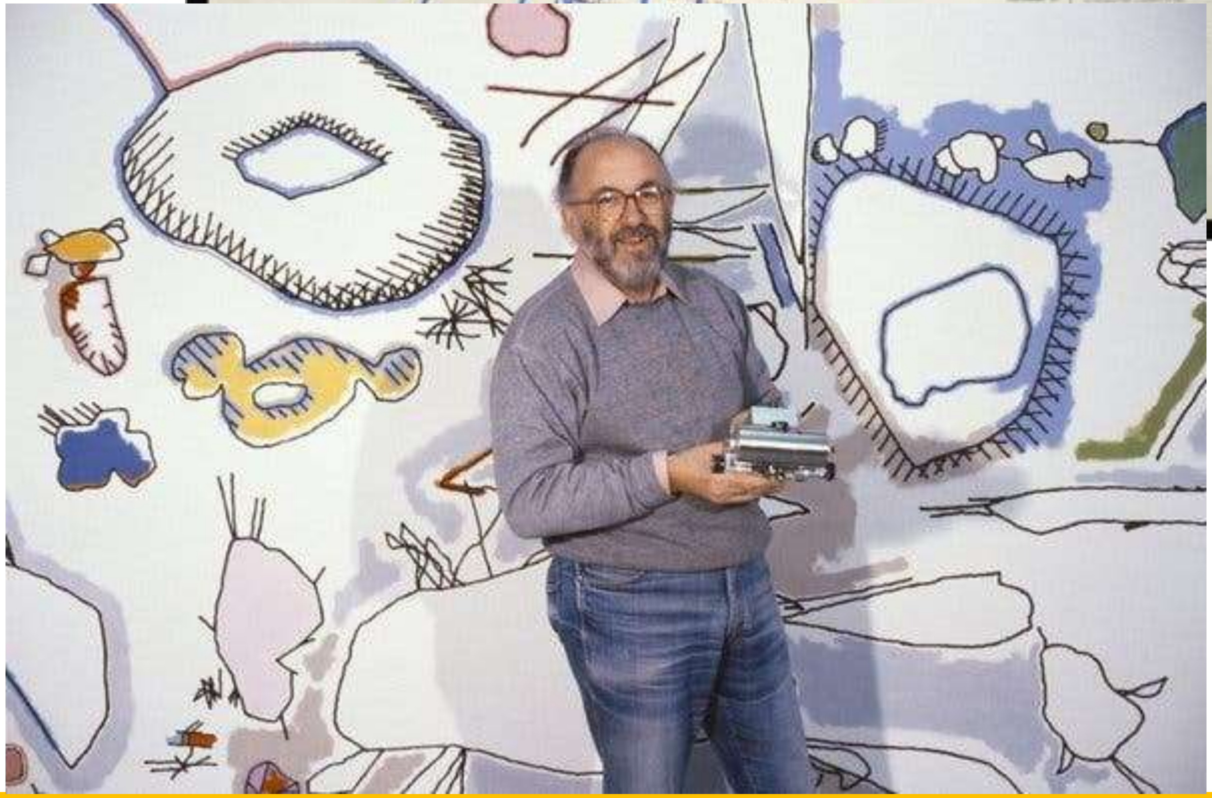
- Computer generated music – who should the credit be given to?
- Can computer be given copyright?
- Can algorithms be given patent?

Earliest debates in AI creativity — whether machines are tools or creators

- Some researchers argue that **AI technologies are creators themselves**, not just tools.
- The debate dates back to **1968 – Cybernetic Serendipity exhibition** curated by **Jasia Reichardt**.
- Early example of AI art:
  - **Harold Cohen's program AARON**
  - Generated images of **plants, people, and abstract forms**
  - Produced artworks resembling **human watercolor paintings**
- Despite the automation, **Harold Cohen was recognized as the creator** and received the **ACM SIGGRAPH Lifetime Achievement Award (2014)**.
- **Core Idea**
- Even when computers generate artwork, **humans are still considered the primary creators**.







- Late Harold Cohen, a Pioneer of Computer-Generated Art

Generative and Interactive Computer Art

- Expanding Human Creativity with Algorithms
- Computer-generated art often includes **geometric and algorithmic patterns**.
- Important contributors:
  - **Paul Brown**
    - Uses **evolving generative patterns**
    - Seeks art “**that makes itself**”
  - **Ernest Edmonds**
    - Creates **interactive art**
    - Artwork **changes depending on the viewer**
- Edmonds emphasizes:
  - Computation **extends and amplifies the creative process**
  - It **does not replace human creativity**
- Both artists’ works are **exhibited globally and collected by major museums**.
- AI and computation are viewed as **creative collaborators**, not replacements for artists.

Paul Brown

Ernest Edmonds

"If an artwork changes depending on the viewer, who is the real creator — the artist, the algorithm, or the audience?"

AI-Generated Images, Music, and Ownership Debate

- Who Owns AI-Generated Creativity?
- Modern AI artists use **machine learning techniques like Generative Adversarial Networks (GANs)**.
- Example:
  - **Alexander Mordvintsev's** DeepDream
  - Produces surreal images such as:
    - distorted animals
    - multiple eyes or heads
    - objects merging with backgrounds

- AI systems can also **generate music and lyrics** in the style of famous musicians.
- Should **credit go to the human programmer** or the **AI algorithm**?
- **Legal Position**
  - **US Copyright Office grants ownership only to humans or organizations.**
  - Algorithms **cannot hold copyright or patents**.
- The major question remains:
  Is AI the creator, or is it simply a tool used by humans?

"If an AI produces something that even the programmer did not expect, who should be credited?"

Will Automation, AI, and Robots Lead to Widespread Unemployment?

1. Weaving machines
2. Printing press
3. Camera/Photography
4. Computers
5. Automatic teller machines

Amazon acquisition of Kiva Robotics in 2012

- **Automation Did NOT Reduce Jobs at Amazon**
- 2012: Amazon acquired Kiva Robotics
- Many predicted large job cuts due to warehouse automation
- Instead, workforce grew from **125,000 to 1+ million (by 2021)**
- COVID-19 accelerated e-commerce demand
- Automation + Demand = More jobs, not fewer
- Did robots replace workers or help Amazon grow faster?
- Amazon created large numbers of jobs
- But:
    - How many small retail jobs were lost?
    - How many supplier/manufacturer jobs were created?
    - Net employment impact remains unclear

- The focus on growth at the cost of poor employee treatment is a serious danger of increased automation.
- **Amazon has fought the introduction of labor unions**, which have in the past enabled workers to get better wages, working conditions, health insurance, and benefits such as child care, sick pay, vacations, and pensions.
- The central question is how to distribute the benefits of automation more equitably.

- **Growth = Opportunities + Controversies**
- **Opportunities**
- Expanded jobs in:
  - Supply chains
  - Fulfillment centers
  - Delivery networks
  - $15 minimum wage (industry-leading at the time)
- **Controversies**
- Reports of demanding work conditions
- Higher injury rates
- Opposition to labor unions

Can a company create jobs and still be criticized?

- The main effect of automation is typically to lower costs and increase quality, which increases demand and expands production, bringing benefits to customers while increasing employment.
- Automation is also disruptive by making certain skills obsolete, so there are substantial challenges to technology innovators and policy-makers, especially in helping those whose jobs are destroyed
- Jobs will be available for those who learn to upskill and learn to use new technology